Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 15-03-2026**
Time: 09.00AM to 12.00PM **GTM-22** **Max. Marks: 300**

KEY SHEET

MATHEMATICS

1	**3**	2	**3**	3	**1**	4	**4**	5	**1**
6	**3**	7	**1**	8	**4**	9	**1**	10	**1**
11	**3**	12	**1**	13	**4**	14	**2**	15	**2**
16	**3**	17	**2**	18	**1**	19	**4**	20	**4**
21	**6**	22	**474**	23	**2**	24	**13**	25	**1**

PHYSICS

26	**4**	27	**1**	28	**1**	29	**2**	30	**4**
31	**1**	32	**3**	33	**3**	34	**2**	35	**3**
36	**1**	37	**2**	38	**1**	39	**1**	40	**1**
41	**2**	42	**1**	43	**4**	44	**1**	45	**4**
46	**60**	47	**5**	48	**30**	49	**40**	50	**500**

CHEMISTRY

51	**3**	52	**3**	53	**4**	54	**4**	55	**4**
56	**3**	57	**1**	58	**4**	59	**3**	60	**4**
61	**3**	62	**3**	63	**2**	64	**3**	65	**4**
66	**2**	67	**3**	68	**1**	69	**3**	70	**1**
71	**5**	72	**1**	73	**4**	74	**2**	75	**2**

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SOLUTIONS

MATHEMATICS

1. $D=\begin{vmatrix}1&1&1\\1&2&3\\1&5&\lambda\end{vmatrix}=\lambda-8\Rightarrow\lambda=8 \quad D_3=\begin{vmatrix}1&1&6\\1&2&14\\2&5&\mu\end{vmatrix}=\mu-36=0\Rightarrow\mu=36$

2. $f\left(0^-\right)=\lim\limits_{x\to0^-}\dfrac{a(1-\sin x)+b\cos x+5}{x^2}=f(0)=\lim\limits_{x\to0^-}\dfrac{(a+b+5)+\left(-a-\dfrac{b}{2}\right)x^2+\ldots}{x^2}=3$

$\Rightarrow a+b+5=0\,and-a-\dfrac{b}{2}=3\Rightarrow a=-1,b=-4\Rightarrow f\left(0^+\right)=\lim\limits_{x\to0^+}\left[1+\left(\dfrac{cx+dx^3}{x^2}\right)\right]^{1/x}$ exits.

$\Rightarrow\lim\limits_{x\to0^+}\dfrac{cx+dx^3}{x^2}=0\Rightarrow c=0$

Now, $\lim\limits_{x\to0^+}(1+dx)^{1/x}=\lim\limits_{x\to0^+}\left((1+dx)^{1/dx}\right)^d=e^d$

So, $e^d=3\Rightarrow d=\log_e 3$

$\therefore a+b+c+d=\log_e 3-5$

3. $f(x)=\dfrac{42^x+16}{2.2^{2x}+16.2^x+32}=f(x)=\dfrac{4\left(2^x+4\right)}{2.2^x+8.2^x+16}=f(4-x)=\dfrac{2}{\left(2^x+4\right)}=f(x)+f(4-x)=\dfrac{1}{2}$

So, $f\left(\dfrac{1}{15}\right)+f\left(\dfrac{59}{15}\right)=\dfrac{1}{2}$

Similarly $=f\left(\dfrac{29}{15}\right)+f\left(\dfrac{31}{15}\right)=\dfrac{1}{2}$

$f\left(\dfrac{30}{15}\right)=f(2)=\dfrac{2}{2^2+4}=\dfrac{2}{8}=\dfrac{1}{4}\Rightarrow8\left(29x\dfrac{1}{2}+\dfrac{1}{4}\right)$

4. (i) Number of numbers created using $1111133=\dfrac{7!}{5!2!}\Rightarrow21$

(ii) Number of numbers created using $1111223=\dfrac{7!}{4!2!}\Rightarrow105$

(ii) Number of numbers created using $1112222=\dfrac{7!}{4!3!}\Rightarrow35$

Total =161

5. General term $={}^nC_r\left\{7^{1/3}\right\}^{n-r}\left(11^{1/12}\right)^r={}^nC_r\left\{7\right\}^{\frac{n-r}{3}}(11)^{r/12}$

For integral terms, r must be multiple of 12

$\therefore r=2k,k\in W$

Total values of r =183

Hence max r =12(182) =2184

Min value of n = 2184

$k=2^3.3^1.7^1.13^1$

6. Points of intersection of $x=\dfrac{11}{2}$ with L_1 & L_3

Gives, $\alpha_{\min}=\dfrac{11}{2}=2\alpha_1$ and $\alpha_{\max}=6=2\alpha_2$

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7. Median $CD = 3r$

$$3r = \frac{1}{2}\sqrt{2(3)^2 + 2(4)^2 - (2r)^2}$$

$$36r^2 = 18 + 32 - 4r^2 \Rightarrow 40r^2 = 50$$

$$r^2 = \frac{5}{4} \Rightarrow r = \frac{\sqrt{5}}{2}$$

Diameter $AB = 2r = \sqrt{5}$

8. $\because f'(x) = ax(x-1) \quad \Rightarrow \quad f'(2) = 6 \quad \Rightarrow \quad a = 3$

$$f'(x) = 3(x^2 - x) \quad \Rightarrow \quad f(x) = x^3 - \frac{3x^2}{2} + C$$

$$\therefore \quad f(x) - x^2\left[x - \frac{3}{2}\right]$$

$$f(2) = 2 \Rightarrow C = 0$$

9. (A) $(1+A)^8 = {}^8C_0 I + {}^8C_1 IA + {}^8C_2 IA^2 + \ldots\ldots\ldots + {}^8C_8 IA^8 = {}^8C_0 I + {}^8C_1 A + {}^8C_2 A^2 + \ldots\ldots\ldots + {}^8C_8 A^8$

$$= I + A\left({}^8C_1 + {}^8C_2 + \ldots\ldots\ldots + {}^8C_8\right) = I + A\left(2^8 - 1\right) \Rightarrow \lambda = 2^8 - 1$$

(B) $\left|\text{adj}\left(A^{-1}\right)\right| = \left|A^{-1}\right|^2 = \frac{1}{|A|^2} = \left|\text{adj}\left(A^{-1}\right)\right| = \frac{1}{\left|\text{adj}A^{-1}\right|} = |A|^2 = 2^2 = 4$

(C) $|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \Rightarrow |B| = \begin{vmatrix} a_{11} & \lambda^{-1}a_{12} & \lambda^{-2}a_{13} \\ \lambda a_{21} & a_{22} & \lambda^{-1}a_{23} \\ \lambda^2 a_{31} & \lambda a_{32} & a_{33} \end{vmatrix} = \frac{1}{\lambda^3}\begin{vmatrix} \lambda^2 a_{11} & \lambda a_{12} & a_{13} \\ \lambda^2 a_{21} & \lambda a_{22} & a_{23} \\ \lambda^2 a_{31} & \lambda a_{32} & a_{33} \end{vmatrix} = |A|$

Hence, $|A| = |B| \Rightarrow 1$

(D) A diagonal matrix is commutative with every square matrix, if it is a scalar matrix. So every diagonal element is 4.

$\therefore |A| = 64$.

10. $\left|\hat{a} + \hat{b} + 2\left(\hat{a} \times \hat{b}\right)\right|^2 = 4$

$$|\hat{a}|^2 + \left|\hat{b}\right|^2 + 4\left|\hat{a} \times \hat{b}\right|^2 + 1\hat{a}.\hat{b} + 4\hat{b}\left(\hat{a} \times \hat{b}\right) + 4\hat{a}\left(\hat{a} \times \hat{b}\right) = 4$$

$$1 + 1 + 4\sin^2\theta + 2\cos\theta = 4$$

$$4\sin^2\theta + 2\cos\theta = 2$$

$$2\sin^2\theta + \cos\theta = 1$$

$$2 - 2\cos^2\theta + \cos\theta = 1$$

$$2\cos^2\theta - \cos\theta - 1 = 0$$

$$2\cos^2\theta - 2\cos\theta + \cos\theta - 1 = 0$$

$$\cos\theta = 1, -1/2$$

$$\theta = 2\pi/3$$

11. $y\frac{dy}{dx} = \frac{y^2 - x}{5x + y^2}$

Let $y^2 = vx$

$$2y\frac{dy}{dx} = v + x\frac{dv}{dx}$$

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$$\frac{1}{2}\left(v+x\frac{dv}{dx}\right)=\frac{vx-x}{5x+vx}$$

$$v+x\frac{dv}{dx}=2\left(\frac{v-1}{5+v}\right)$$

$$x\frac{dv}{dx}=-\frac{v^2+3v+2}{5+v}$$

$$\int\frac{5+v}{(v+1)(v+2)}dv=-\int\frac{dx}{dx}$$

$$\int\left(\frac{4}{v+1}-\frac{3}{v+2}\right)dx=-\int\frac{dx}{x}$$

$$4\log|v+1|-3\log|v+2|=\log x+\log c$$

12. Let $I=\int_{\log\sqrt{\frac{\pi}{2}}}^{\log\sqrt{x}}e^{2x}\sec^2\left(\frac{1}{3}e^{2x}\right)dx=\frac{3}{2}\tan\left(\frac{1}{3}e^{2^x}\right)_{\ln\sqrt{\frac{\pi}{2}}}^{\ln\sqrt{\pi}}=\frac{3}{2}\left[\tan\frac{\pi}{3}-\tan\frac{\pi}{6}\right]=\sqrt{3}$

13. $n_1=50,\bar{x}_1=630,\sigma_1=90$

$n_2=40,\bar{x}_2=540,\sigma_2=60$

$$\sigma^2=\frac{n_1\sigma_1^2+n_2\sigma_2^2}{n_1+n_2}+\frac{n_1n_2}{(n_1+n_2)^2}(\bar{x}_1-\bar{x}_2)^2$$

14. $3x^2=x^2+18$

$x=\pm3$

Area between P_1 & P_2

$$=\int_{-3}^{3}\left(x^2+18-3x^2\right)dx=\int_{-3}^{3}\left(18-2x^2\right)dx.$$

$$=2\int_0^3\left(18-2x^2\right)dx=72$$

Area bounded between P_1 & $line=6\,sq\ units$

Area bounded by $x^2=4ay$ 7 $x=my$ *is* $\frac{8a^2}{3m^3}$

$\therefore$ *Area bounded by* $x^2=\frac{y}{3}$ & $x=\frac{y}{k}$

$$is\ \frac{8\left(\frac{1}{9}\right)}{3.\left(\frac{1}{k^3}\right)}=6$$

15. $2|z+3i|=|z-i|$

Put $z=x+iy$ $\Rightarrow 2|x+(y+3)i|=|x+i(y-1)|$

$$\Rightarrow 2\sqrt{x^2+(y+3)^2}=\sqrt{x^2+(y-1)^2}$$

$$\Rightarrow 4\left(x^2+(y+3)^2\right)=x^2+y^2+\frac{26y}{3}+\frac{35}{3}=0$$

$$C=\left(0,-\frac{13}{3}\right),r=\sqrt{\frac{169}{9}-\frac{35}{3}}=\sqrt{\frac{169-105}{9}}=\frac{8}{5}\qquad\therefore g+f-r=0-\frac{13}{3}-\frac{8}{3}=-7$$

16. Ellipse foci are $(2,0),(-2,0)\Rightarrow f_1=2,f_2=-2$

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Parabola P_1 has vertex $(0,0)$, focus $(2,0)$, directrix x=-2

P_2 has $V(0,0)$, focus $(-4,0)$, directrix x=4

17. $x^2-25\geq 0\Rightarrow x\in[-\infty,-5)\cup[5,\infty)$ and $4-x^2\neq 0\Rightarrow x\neq\pm 2$ and $x^2+2x-15>0\Rightarrow(x+5)(x-3)>0\Rightarrow x\in(-\infty,-5)\cup(3,\infty)$ so, the domain will be $(-\infty,-5)\cup[5,\infty)$

18. $\lim\limits_{x\to a^+} g(x)=g(a)=0, \lim\limits_{x\to a_-} g(x)=0\Rightarrow g(x)$ is continuous at x=a.

$$\lim_{x\to a^-}\frac{g(x)-g(a)}{x-a}=0, \lim_{x\to a^+}\frac{g(x)-g(a)}{x-a}=\lim_{x\to a^+}\frac{\int_a^x f(t)\,dt}{x-b}=f(a)>1$$

ANs $\lim\limits_{x\to a^-} g(x)=g(b)=\int_a^b f(t)\,dt, \lim\limits_{x\to b^+} g(x)=\int_a^b f(t)\,dt\Rightarrow g(x)$ is continuous at x=b

$$\lim_{x\to b^-}\frac{g(x)-g(a)}{x-a}\lim_{x\to b^-}\frac{\int_a^x f(t)\,dt-\int_a^b f(t)\,dt}{x-b}=f(b)>1$$

$$\lim_{x\to b^+}\frac{g(x)-g(a)}{x-a}=0$$

19. Equation of the ellipse is $\frac{x^2}{9}+\frac{y^2}{1}=1$

Equation of auxiliary circle is $x^2+y^2=9$.....(1)

Equation of line AB is $\frac{x}{3}+\frac{y}{1}=1\Rightarrow x=3(1-y)$

Putting this in (1), we get $9(1-y)^2+y^2=0\Rightarrow 10y^2-18y=0\Rightarrow y=0,\frac{9}{5}$

Thus, y coordinate of 'M' is $\frac{9}{5}$

$$\Delta OAM=\left(\frac{1}{2}\right)(OA)(MN)=\frac{1}{2}(3)\frac{9}{5}=\frac{27}{10}$$

20. A= numbers which are divisible by 2 $\Rightarrow n(A)=1011$

B= numbers which are divisible by 3 $\Rightarrow n(B)=674$

C= numbers which are divisible by 337 $\Rightarrow n(C)=6$

$A\cap B$ = numbers which are divisible by 6 $\Rightarrow n(A\cap B)=337$

$A\cap C$ = numbers which are divisible by 674 $\Rightarrow n(A\cap C)=3$

$B\cap C$ = numbers which are divisible by 981 $\Rightarrow n(B\cap C)=2$

$A\cap B\cap C$ = Numbers which are divisible by 2022 $\Rightarrow n(A\cap B\cap C)=1$

$n(A\cup B\cup C)=1350$

21. points satisfying the inequalities are common to all 3 circles, given by $(1,1),(1,2),(2,1),(2,2),(2,3,),(3,2),$

22. $s+5d=8$

$a=8-s\frac{3}{2}=\frac{1}{2}$

Roots are

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$T_{10} = a + 9d = \frac{1}{2} + 9x\frac{3}{2} = 14$

$T_{10} = a + 10d = \frac{1}{2} + 10x\frac{3}{2} = \frac{31}{2}$

$\frac{p}{3} = T_{10} + T_{11} = 14 + \frac{31}{2} = \frac{59}{2}$

$p = \frac{177}{2}$

$\frac{q}{3} = T_{10} \times T_{11} = 7 \times 31 = 217$

$q = 651$

$q - 2p = 651 - 177 = 474$

23. $|\bar{c} + \bar{d}| = \sqrt{21}$

$let\, b = 4\hat{i} - \hat{j} + 2\hat{k}$

$\bar{c} \times \bar{b} = \bar{b} \times \bar{d}$

$(\bar{c} + \bar{d}) \times \bar{b} = 0$

$\bar{c} + \bar{d} = +\bar{b}$

$|\bar{c} + \bar{d}| = 1 + 1\sqrt{16 + 1 + 4}$

$t = \pm 1$

$(\bar{c} + \bar{d}).(\hat{i} - 5\hat{j} - 4\hat{k}) = \pm(4 + 5 - 8) = \pm 1$

$\lambda_1 = 1, \lambda_2 = -1$

$(2\lambda_1 - k)x^2 + (k^2 + 5k - 6\lambda_2)xy + k^2y^2 - 6x - 8y - 3\lambda_2 = 0$

$represents\ a\ circle\ k^2 + 5k - 6\lambda_2 = 0 \Rightarrow k = -2\ or\ -3$

$2\lambda_1 - k = k^2 \Rightarrow k^2 + k - 2\lambda_1 \Rightarrow k = -2 or 1$

$Hence\ k = -2 \Rightarrow -k = 2$

24. The given line: $\frac{x+6}{-8} = \frac{y-7}{12} = \frac{z+5}{-16} = \lambda$

$Q = (2\lambda - 6, 7, -3\lambda, 4\lambda - 5)$

Now, $\overrightarrow{QR} = (2\lambda - 7, -3\lambda, 4\lambda - 11)$

Since, QR is perpendicular to the line

$\therefore QR(dr's\ of\ line) = 0$

$\Rightarrow 4\lambda - 14 + 9\lambda + 16\lambda - 44 = 0$

$\Rightarrow 29\lambda = 58 \Rightarrow \lambda = 1$

$\therefore Q = (-2, 1, 3)$

$PQ = \sqrt{144 + 9 + 16} = \sqrt{169} = 13$

25. $f''(x)\begin{cases} a^2xe^aex + 2ae^{ax} & : x \le 0 \\ 2a - 6x & : x > 0 \end{cases}$

$f'(x)$ Increasing if $(ax + 2)ae^{ax} \ge 0$ and $2a - 6x \ge 0$

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PHYSICS

26. *Use*

$\phi = BA$

$B = \frac{\mu_0 l}{2R}$

$[B] = ML^0T^2A^{-1}$

$[L] = ML^2T^{-2}A^{-2}$

27. Statement-I is true but Statement-II is true

28. $B = \frac{\mu_0 i}{2\pi r}$

29. Area under P-V diagram gives work done by gas.

30. Both (A) and (R) are false

31. $KE = \frac{1}{4\pi \in_0} \frac{q_1 q_2}{r}$

32. $\vartheta = 3\sqrt{x}$

$a = \vartheta \frac{d\vartheta}{dx}$

$a = 3\sqrt{x} \frac{1}{25x}$

$= \frac{3}{2}$

33. $dw = \frac{m}{L} gx \sin\theta dx$

$\frac{1}{2}mv^2 = \int_b^l dw$

34. by symmetry moment of inertia is equal

35. $\frac{1}{2}mv^2 - \frac{GMm}{R} = -\frac{GMm}{R+h}$

36. $v = \sqrt{2g(H-h)}$

$R = \sqrt{2g(H-h)}\sqrt{\frac{2(h+H)}{g}}$

37. K.E=$\frac{3}{2}KT$ no of molecules

38. $f_c = \frac{3v}{4L_1}$

$f_0 = \frac{V}{L_2}$

39.

40. $E_0 = \frac{kp_0}{R^3}$

$R = \left(\frac{Kp_0}{E_0}\right) y_3$

41. $r = \frac{mv}{bq} \Rightarrow r = \frac{\sqrt{2mKE}}{bq}$ $E = \rightarrow \left[\frac{L}{x^2} - \frac{L}{x^2}\right]$

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42. $M = \frac{loNN_2A}{L}$

$e = -\frac{mdi}{dt}$

43. $\frac{d\theta}{dt} = \left(\frac{d\theta}{dt}\right)_1 + \left(\frac{d\theta}{dt}\right)_2$

44.

$f_0 + f_e = 30$

$\frac{f_0}{f_e} = 2$

$then\ f_0\ is\ 20$

45. $c = \frac{\varepsilon A}{d}$

$c^1 = \frac{\varepsilon A^1}{d^{-1}}$

46. Combination of lens

47. $S_{1P} - S_{2P} = \frac{d^2}{2D}$

$= \frac{5}{2}d$

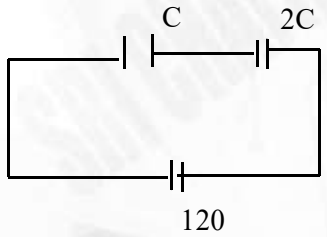
48.

$\frac{6}{l} = \frac{4}{50-l}$

$300 - 6l = 4l$

$l = 30cm$

49.



$V_{2c} = \frac{\theta}{2C} = \frac{V}{2c}$

50. $p.f = \cos\theta = \frac{R}{Z}$

$Z = \sqrt{x^2 + (x2 - xc)^2}$

$Z = \sqrt{2}R$

$Z = 10\sqrt{2}$

$c = 500\mu F$

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CHEMISTRY

51. Diphenylbenzene-1,3-dicarboxylate

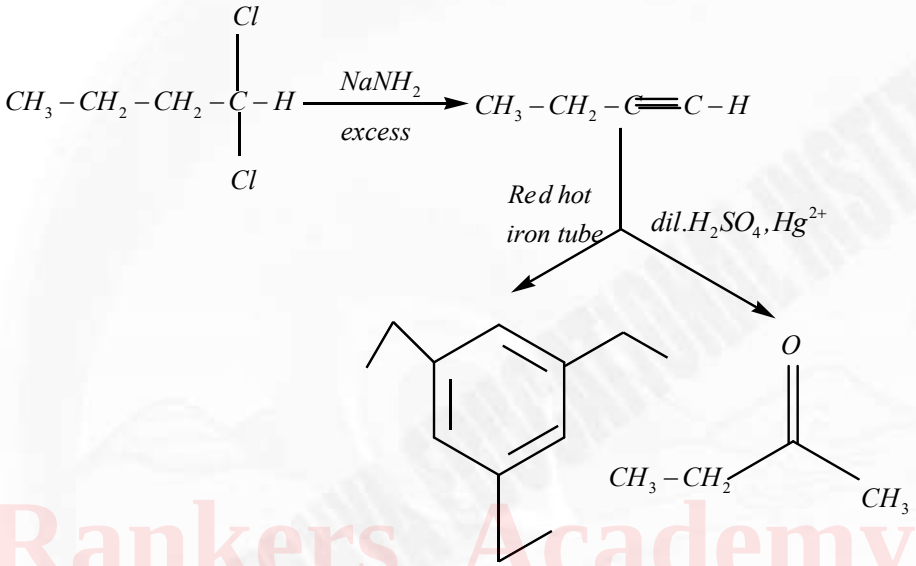
52.

$\overset{\ominus}{C}F_3 < \overset{\ominus}{C}Cl_3$

$(p\pi - d\pi)$ bond

53. $c < d < f < a < b < e$

54.

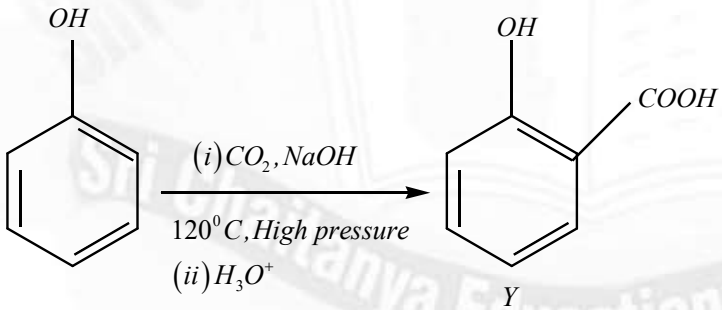


55.

$2 \times V.D = Molar\ Mass$

$2 \times 47 = 94$

Gives colour with neutral $FeCl_3$. *Compound is phenol*



Both X and Y are soluble in NaOH

Y will dissolve in $NaHCO_3$ and evolve CO_2 gas

Both X and Y being aromatic will burn with sooty flame.

However salicylic acid is more acidic than phenol.

56. Hofmann's elimination, Fluorine is Poor Leaving Group

57. Average rate depends upon the change in concentration of reactants or products and the time taken for that change to occur. Average rate cannot be used to predict the rate of a reaction at a particular instant as it would be constant for the time internal for which it is calculated. So, to express the rate at a particular moment of time, we determine the

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instantaneous rate. It is obtained when we consider the average rate at the smallest time interval say dt(i.e. when it approaches zero).

58. Only solvent molecules can passed through SPM so, only dilution is possible.

59. Given $In^{3+} + 2e^- \rightarrow In^+; E_1^\circ = -0.42V .....(i)$

$In^{2+} + e^- \rightarrow In^+; E_2^\circ = -0.40V ....(ii)$

By subtracting Eq. (ii) from Eq. (i) a third half-cell reaction can be obtained as:

$$\left[\begin{array}{l} Where\ E_3^\circ \times 1 \times F = E_1^\circ \times 2 \times F - E_2^\circ \times 1 \times F \\ or \qquad E_3^\circ = 2 \times (-0.42) - 1 \times (-0.40) = -0.44V \end{array}\right]$$

$In^{3+} + e^- \rightarrow In^{2+}; E_3^\circ = -0.44V$

For the reactions: $Cu^{2+} + e^- \rightarrow Cu^+; E_4^\circ = 0.15$

$In^{2+} \rightarrow In^{3+} + e^-; E_3^\circ = +0.44$

The net redox change: $Cu^{2+} + In^{2+} \rightarrow Cu^+ + In^{3+};$

$$E_{cell}^\circ = E_4^\circ + E_3^\circ = 0.15 + 0.44 = 0.59V$$

Also $E_{cell}^\circ = \dfrac{0.059}{1} \log K_c$

$0.59 = \dfrac{0.059}{1} \log K_c \quad \therefore K_c = 10^{10}$

60. Synergic bond

61. $F^- - WFL,\ C_2O_4^{-2} SFL$

62. $Acidic - CO_2, GeO_2, SiO_2; Neutral - CO$

$Amphoteric - SnO_2, SnO$

63. $A = NH_4NO_2, B = NO_2; C = NH_3; D = N_2$

64. When both electric and magnetic field is applied, electron strikes at point B, and at point C when only magnetic field is applied.

65. ΔH needed for change $= \Delta H_{heating} + \Delta H_{vaporisation}$

$= 50 \times 10^3 \times 1 \times (100 - 10)^\circ C + 540 \times 50 \times 10^3$

$= 31500 \times 10^3 cal$

ΔH actually needed $= \dfrac{31500 \times 10^3 \times 100}{60}$

(since only 60% of heat is used to do so)

Now $368 \times 10^3 cal$ heat is given on combustion by $22.4\ litre\ C_2H_6$

$\therefore \dfrac{31500 \times 10^3 \times 100}{60}$ heat is given by

$= \dfrac{22.4 \times 31500 \times 10^3 \times 100}{60 \times 368 \times 10^3}$

$= 3.196 \times 10^3 litre = 3.196 m^3$

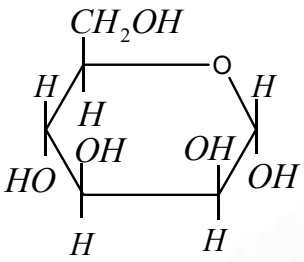
66. Lower the total energy, more is the stability of the ion/molecule.

H_2^+ is more stable than He_2^+

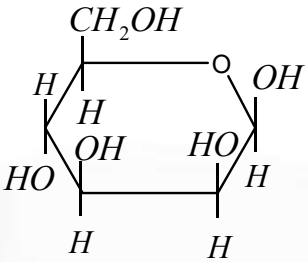
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67. PIC

68. The structure of D-Mannose is

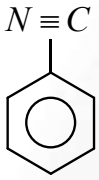


$(\alpha - form)$



$(\beta - form)$

69.



70. A) $p_z + p_y \Rightarrow$ Non-bonding if inter nuclear axis is x-axis

B) $p_y + p_x \Rightarrow$ Form non-bonding if inter nuclear axis is x-axis

C) $d_{xz} + d_{xz} \Rightarrow$ can form δ – bond if inter nuclear axis is y-axis

D) $p_z + p_y \Rightarrow$ can not form π – bond, if inter nuclear axis is x-axis

71.

$A \rightarrow B$

$K_{560} = 2K_{300}$

$t_{\underset{500}{50\%}} = 2$

$K_{500} = \dfrac{0.693}{2}$

$K_{300} = \dfrac{0.693}{4}$

$C \rightarrow D$

$E_{a_{C\rightarrow D}} = E_{a_{\frac{A\rightarrow B}{2}}} = \dfrac{519.90R}{2} = 259.95R$

$C \rightarrow D$

$K_{500_{C\rightarrow D}} = 2 \times K_{500_{A\rightarrow B}}$

$= 2 \times \dfrac{0.693}{2}$

$= 0.693$

$log\dfrac{K_{500}}{K_{300}} = \dfrac{E_a}{2.303R}\left(\dfrac{1}{300} - \dfrac{1}{500}\right)$

$log\left(\dfrac{0.693}{K_{300}}\right) = \dfrac{259.95R}{2.303R}\left(\dfrac{200}{300\times 500}\right)$

$K_{300} = 5$

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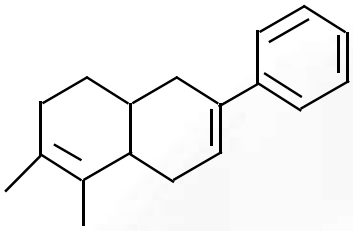
72. $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

$3Sn^{2+} - 6e^- \rightarrow 3Sn^{4+}$

Hence $Cr_2O_7^{2-} + 14H^+ + 3Sn^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 3Sn^{4+}$

$\therefore$ 3 moles of Sn^{2+} reduces 1 mole of $K_2Cr_2O_7$.

73.



74. $\pi = CRT = \rho gh$

$$\frac{0.2/M}{100/1000} \times 0.0821 \times 300 = \frac{1.013 \times 0.2463}{1.013 \times 10^6}$$

$$\therefore M = 2 \times 10^5$$

75.

$$\lambda_d = \frac{h}{\sqrt{2mK.E}}$$

Here KE is same i.e, 400KeV

$$so\ \lambda_d \propto \frac{1}{\sqrt{m}}$$

$$\frac{(\lambda_d)_{m_1}}{(\lambda_d)_{m_2}} = \sqrt{\frac{m_2}{m_1}} = \sqrt{4} = 2$$

$$(\lambda_d)_{m_1} = 2(\lambda_d)_{m_2}$$

$$so\ x = 2$$

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